Assignment 8: NLP Pipeline and Text Classification

NorthStar Telecom – Customer Support Message Classification

Course:	**MBAI 5310G – AI Programming**
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Dataset:	NLP_Dataset_2.xlsx — NorthStar Telecom Service Complaint Classification

1. Business Problem

NorthStar Telecom receives hundreds of customer support messages daily across five channels: Email, Chat, Call Center, Mobile App, and Web Form. Each message must be routed to the appropriate support team — Billing, Internet Service, Mobile App, Technical Support, Account Access, or Cancellation. Manual routing is slow and error-prone.

This assignment builds an NLP-based text classification model that reads the raw text of each incoming ticket and automatically predicts the correct IssueCategory, enabling faster routing and improved customer experience.

2. Dataset

The dataset contains 120 synthetic NorthStar Telecom customer support tickets. It is perfectly balanced, with exactly 20 records per category, and contains no missing values.

Column	Description
TicketID	Unique ticket identifier (e.g., TCK-001) — not used as a model feature
TicketDate	Date the message was received
Source	Channel: Email, Chat, Call Center, Mobile App, or Web Form
City	Customer city (e.g., Ottawa, Toronto, Brampton)
PlanType	Subscription tier: Basic, Plus, Family, Business, or Student
CustomerType	Residential, Small Business, Student, or Remote Worker
MessageText	Raw customer support message — NLP input feature
IssueCategory	Target label — six-class classification
Priority	Ticket urgency: Low, Medium, or High

3. NLP Preprocessing Pipeline

The raw MessageText column was preprocessed through the following five steps before feature extraction:

Step	Technique	Purpose
1	Lowercasing	Normalise text — 'Crash' and 'crash' treated as the same token
2	Punctuation removal	Strip non-alphabetic characters using regex [^a-z\s]
3	Tokenization	Split text into individual tokens using nltk.word_tokenize
4	Stopword removal	Remove filler words (NLTK English list) that carry no category signal
5	Lemmatization	Reduce words to base form — 'crashes' → 'crash', 'received' → 'receive'

After preprocessing, a CleanedText column was created containing the normalised, tokenised, lemmatized text ready for vectorization.

4. Exploratory Text Analysis

Word frequency analysis was performed on the full cleaned corpus. The most common words across all 120 tickets were:

northstar (30), still (30), plan (25), app (20), service (19), please (18), ticket (16), created (15), alex (15), modem (13), account (12), internet (11)

Two groups of words emerge from this list:

- Meaningful category signals: app, modem, account, internet, service — these terms map directly onto specific issue categories and are the primary drivers of classification.
- Template artefacts: alex, ticket, created — these appear because many messages in the synthetic dataset contain a repeated boilerplate phrase ('Ticket created by Alex'). They carry no complaint-type signal and would be removed by a domain-specific stopword list in a production system. TF-IDF's inverse document frequency weight partially suppresses them since they appear across all six categories.

5. POS Tagging and Named Entity Recognition

Part-of-speech tagging and named entity recognition were applied to three representative tickets from different categories.

POS Tagging Findings

- Nouns are the strongest classification signals — app and plan in the Mobile_App ticket; price, notice, and fee in the Billing ticket; router and light in Technical_Support.
- Verbs describe the nature of the complaint — crashes and fails vs. increased vs. connect — and help distinguish categories that share similar nouns.
- Adjectives add specificity — mobile (app type), monthly (billing period), red (hardware state).

Named Entity Recognition Findings

- NorthStar is correctly identified as an ORGANIZATION entity across all tickets.
- City names (Ottawa, Toronto) are recognised as LOCATION entities where mentioned.
- In a production pipeline, NER could auto-extract city for regional routing, dates for time-sensitive flagging, and agent names to link tickets to case history.

6. Feature Extraction (TF-IDF)

TF-IDF (Term Frequency – Inverse Document Frequency) was used to convert the preprocessed text into a numerical matrix. The vectorizer was configured with max_features=500 and ngram_range=(1, 2), capturing both individual words and two-word phrases (e.g., mobile app, internet service, account access).

Critically, the TF-IDF vectorizer was fit only on the 96 training samples and then applied (transform only) to the 24 test samples. Fitting on the full dataset would leak information from test records into training, artificially inflating performance metrics. The final matrix contained 376 features.

7. Classification Model

Logistic Regression was selected as the classification algorithm. It is a fast, interpretable linear classifier that performs well on TF-IDF features, learning a separate weight per term per category so that predictions can be traced back to specific words.

Parameter	Value	Rationale
Algorithm	Logistic Regression	Strong baseline for text classification with TF-IDF features
Train / Test split	80% / 20% (96 / 24 samples)	Standard split; stratify=y ensures proportional class representation
max_iter	1000	Allows convergence on a multi-class problem
C (regularisation)	1.0	Default regularisation; prevents overfitting
random_state	42	Ensures reproducibility

8. Model Evaluation

Results

Metric	Score
Accuracy	100%
Precision (macro avg)	1.00
Recall (macro avg)	1.00
F1-Score (macro avg)	1.00
Test samples	24 (4 per category)

Critical Interpretation

100% accuracy is expected for a synthetic dataset with template-based, category-specific vocabulary. This result should not be interpreted as a guarantee of production performance. Real customer messages contain spelling errors, abbreviations, mixed topics, and informal language that a model trained on this data would struggle to classify correctly. Realistic accuracy on real NorthStar Telecom messages would be approximately 75–90%, depending on data volume and quality.

Precision vs. Recall Trade-off

For this business use case, recall is the more critical metric. A missed classification (low recall) means a ticket is unrouted or routed to the wrong team, causing resolution delays and risking customer churn — particularly for Billing and Account_Access where unresolved issues have direct financial consequences. A false alarm (low precision) is also harmful but is easier to detect and correct downstream.

In a production deployment, per-category classification thresholds could be tuned independently to maximise recall for high-priority categories, accepting slightly lower precision where the cost of missed tickets is highest.

Confusion Matrix

All 24 test tickets fell on the main diagonal — zero misclassifications on this synthetic dataset. In a real deployment, the most likely error pairs would be Technical_Support and Internet_Service (both involve connectivity language) or Account_Access and Cancellation (both involve account-level actions). These misroutes cause a small delay but are not severe errors.

9. Business Interpretation

Item	Detail
What the model predicts	IssueCategory — six-class classification of incoming NorthStar Telecom support tickets
Dataset used	NLP Dataset 2 — NorthStar Telecom, 120 synthetic tickets, 20 per category
Text column	MessageText (raw customer message)
Target column	IssueCategory
Preprocessing	Lowercase → punctuation removal → tokenize → stopwords → lemmatize → TF-IDF (unigrams + bigrams)
Model	Logistic Regression, 80/20 stratified split
Result	100% accuracy on 24 test samples — proof-of-concept; realistic real-world accuracy: 75–90%
Business insight	Template artefacts (repeated boilerplate phrases) inflate frequency of non-signal words; domain-specific stopword filtering needed for production
Limitation	Synthetic data with controlled vocabulary — real messages require retraining on labelled production data

10. Responsible AI Reflection

Potential Sources of Bias

- Channel bias: Customers using Call Center may produce transcribed speech with different language patterns than typed messages from Chat or Email channels. A model trained on mixed-channel data may perform unevenly across customer groups that prefer specific channels — for example, elderly customers or those with limited digital literacy who predominantly call rather than type.
- Language and literacy bias: The model was trained on well-formed English sentences. Customers who write in fragmented English, use heavy abbreviations, or mix languages may receive lower classification accuracy, leading to slower service for those groups — disproportionately affecting non-native English speakers.
- City and plan-type representation: If the training data over-represents certain cities or plan types, the model may be less accurate for customers in underrepresented regions or on less common subscription tiers.

Human Oversight Requirement

This model should support human agents, not replace them. The recommended approach is a confidence-threshold system: tickets where the model's predicted probability exceeds 90% are auto-routed to the appropriate team; tickets with lower confidence are flagged for manual review by a supervisor. This ensures that ambiguous or unusual messages always receive human attention.

Model performance should be monitored regularly by category and by customer segment to detect drift or disparate error rates before they affect service quality. The model's outputs should never be used as a basis for denying service to customers — classification is a routing aid, not an eligibility decision.

Data Privacy

Customer support messages may contain personally identifiable information (PII), including names, account numbers, addresses, and payment details. Any production NLP pipeline must include PII detection and redaction before data is used for model training or retraining to comply with privacy regulations.

11. Conclusion

This assignment demonstrates a complete NLP pipeline applied to a text classification problem in the telecommunications sector. The pipeline covers all stages from raw text ingestion to model evaluation: preprocessing (lowercasing, punctuation removal, tokenization, stopword removal, lemmatization), feature extraction (TF-IDF with bigrams), exploratory analysis (word frequency, POS tagging, NER), and classification (Logistic Regression with a stratified train/test split).

The model achieves 100% accuracy on the synthetic NorthStar Telecom dataset, demonstrating that NLP-based ticket routing is technically feasible. Even at a more realistic 80–85% accuracy on real customer data, automating the routing of the majority of incoming tickets would significantly reduce manual workload, improve response times, and allow support staff to focus on genuinely complex or ambiguous cases.

To progress toward production deployment, the next steps are: (1) collect and label real NorthStar Telecom support tickets; (2) retrain the model on this data with domain-specific preprocessing; (3) implement a confidence-threshold routing system with human review for low-confidence predictions; and (4) monitor model performance across channels and customer segments to detect and address bias or drift.